Scoring, substitution matrices and score aggregation for TCR antigen-specificity search

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Appendix A — vdjmatch technical appendix

Abstract

`vdjmatch` annotates T-cell receptor (TCR) clonotypes by antigen specificity through fuzzy CDR3 search against a labelled database (VDJdb), built on the `seqtree` search core. This note documents the *scoring* layer: how CDR3 similarity is scored, the data-driven TCR substitution matrix (VDJAM) and why it is region-structured, and how per-hit scores are aggregated into a control-calibrated significance. CDR3 loops resist the assumptions behind BLOSUM/PAM — they cannot be multiply aligned across clonotypes of different length, and germline-encoded flanks (`CASS`...) are so over-conserved that a naive matrix conflates germline bias with selection. We give a position-debiased log-odds estimator that isolates the substitution signal, show empirically that this signal lives in the non-template (NDN) core while the V/J flanks are germline-determined, and demonstrate the consequences on worked examples (GIL, NLV) using the alignments and CIGAR strings produced by `seqtree`. The E-value theory used for significance is derived in the companion `seqtree` appendix; here we summarize it and give the paired-chain extension.

A.1. The search-and-score framework

Given a query CDR3 q and a labelled reference set D (VDJdb), `vdjmatch` retrieves the neighbours of q within a fixed scope $\theta = (s, i, d, t)$ (max substitutions, insertions, deletions, total edits) using the `seqtree` engine, which defines a ball $B_\theta(q) = \{x : \mathrm{edit}(q, x) \leq \theta\}$ and returns, per hit, the edit decomposition and a matrix-weighted alignment. The matrix enters as a *ranking* score inside the ball; the ball itself (an edit budget) bounds the candidate set. Each hit carries its antigen labels, an extended CIGAR (=/`X`/`I`/`D`), and an alignment score. Significance comes from comparing the neighbour count to a matched background control (§ A.3).

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A.2. Substitution scoring and the VDJAM matrix

How BLOSUM is built, and why it does not transfer.. BLOSUM is a log-odds matrix estimated from the BLOCKS database of *ungapped multiple-alignment blocks* of conserved protein regions. Its recipe is: (i) take many trusted blocks — columns of homologous, gaplessly aligned residues; (ii) cluster sequences within a block above a percent-identity threshold and down-weight within-cluster pairs so abundant near-duplicates do not dominate; (iii) count, over all blocks, how often residues a and b co-occur in the *same alignment column*, q_{ab}; (iv) score $s(a,b) = \frac{1}{\lambda}\log(q_{ab}/p_a p_b)$ against the *global* background frequencies p_a. The signal is "which residues substitute for one another within conserved, column-homologous positions," and the background is position-independent.

CDR3 loops break every step. (i) There is no trusted multiple alignment: same-specificity CDR3s differ in length and have no column homology, so "alignment columns" do not exist. (ii) The germline-encoded ends are over-conserved — essentially every human TRB CDR3 begins `CASS` and ends in a short J motif — so a *global* background p_a is wrong: an alanine in the N-terminal `CASS` is germline-fixed and carries no specificity signal, while an alanine in the centre does, yet BLOSUM would score them identically. A single global matrix estimated over all positions thus conflates germline conservation with antigen-driven selection.

What VDJAM changes (point by point).. VDJAM keeps the log-odds idea but replaces the two broken ingredients. (a) *Observation unit*: instead of columns of a multiple alignment, the unit is a *same-antigen single-substitution pair* — two CDR3s annotated to the same epitope, equal length, differing at exactly one position; this is the only "aligned residue pair" obtainable without a multiple alignment, and (like a BLOCKS column pair) it records two residues that are interchangeable without changing function (here, specificity). (b) *Background*: instead of a global p_a, a *position-specific* background $\mathrm{bg}(\cdot \mid L, p)$ (the residue frequency at that length and position) is used, so over-conserved columns are normalized away analytically rather than masked by hand. (c) *Redundancy*: BLOSUM's percent-identity clustering is replaced — partly by the position-specific background (germline-shared residues are "expected" and contribute little), and prospectively by the control-calibrated significance layer (§ A.3) that already discounts convergent public clones; per-clonotype clustering can be added exactly as in BLOSUM if needed. (d) *Scope*: BLOSUM is one global matrix; VDJAM is estimated per chain and per region, because (next paragraph) only the NDN core carries a transferable substitution signal.

Observation unit and estimator.. We replace "alignment columns" with *same-antigen single-substitution pairs*: ordered pairs of CDR3s annotated to the same epitope, of equal length, differing at exactly one position — the only clean substitution signal extractable without a multiple alignment. For a substitution $a \leftrightarrow b$ observed at position p in a CDR3 of length L on chain c, let $\mathrm{bg}(\cdot \mid L, p)$ be the residue frequency at position p among all length-L chain-c CDR3s. With $f(a,b)$ the observed count of $\{a,b\}$ events and

$$e(a,b) = \sum_{\text{event slots } (L,p)} n_{L,p}\left[\mathrm{bg}(a \mid L,p)\,\mathrm{bg}(b \mid L,p) + \mathrm{bg}(b \mid L,p)\,\mathrm{bg}(a \mid L,p)\right], \qquad \mathrm{VDJAM}(a,b) = \log_2 \frac{f(a,b)+\alpha}{e(a,b)+\alpha}, \tag{1}$$

where $n_{L,p}$ is the number of observed events at (L, p) and α a Laplace pseudocount. The key point is that the background is *position-specific*: at an over-conserved column the residue frequency is peaked, so substitutions there are both rare (small f) and "expected" (matched e), and contribute ≈ 0 to the log-odds — the germline bias is removed analytically rather than by hand-masking. The matrix is consumed by `seqtree` via the squared-distance penalty $\mathrm{pen}(a, b) = s_{aa} + s_{bb} - 2s_{ab}$ (`SubstitutionMatrix.from_similarity`); a dominant uniform diagonal makes higher learned similarity rank a substitution better.

Region structure: NDN is learned, V/J are germline-determined.. A CDR3 is V-germline prefix · NDN insert · J-germline suffix. Estimating (1) separately by region shows (Fig. 1) that the substitution signal — the agreement with general amino-acid chemistry, measured as correlation with BLOSUM62 — is concentrated in the NDN core (TRB $r = 0.42$, TRA 0.35), whereas the V flank carries almost none (TRB 0.08, TRA 0.07): V-flank "substitutions" are dominated by germline allele differences, not free chemistry. This motivates a two-part model: **VDJAM is learned for the NDN core**, while the **V and J flanks are auto-computed** from the germline V/J sequences and their trimming. Concretely, a CDR3 position derived from V- (or J-) germline is near-invariant when it survives trimming and reverts to the random-insert background when trimmed; so the V/J contribution is a per-position conservation profile $w(p) = \Pr[\text{position } p \text{ is germline-retained}]$, determined by the gene's trimming-length distribution, rather than a substitution matrix learned from data. Combined, the NDN matrix scores the core and $w(p)$ down-weights substitutions at germline-retained flanks (realized through a per-position weighting of the base matrix). The germline-retention profile $w(p)$ is computed from the OLGA recombination model (per-gene V/J deletion distributions and germline CDR3-portion segment lengths, via `mirpy`'s `mir.basic.trimming`) and bundled with `vdjmatch`; Fig. 2 shows it drops monotonically from each anchor (the `CASS` V-stem and the J motif are germline; the core is insert). Region-aware scoring weights each CDR3 column by 1 — Pr[germline] and applies the NDN matrix — implemented in `vdjmatch.match.regions` and evaluated below.

A.3. Score aggregation and significance

Per-hit to per-query.. A region-aware similarity score sums the NDN matrix score over the core and the (down-weighted) flank contributions; this is available as an optional ranking score and generalizes the legacy cloglog generalized-linear aggregation over V/CDR1/CDR2/J/CDR3.

Control-calibrated E-value (primary).. Immune repertoires are biologically redundant, so an i.i.d. null over-calls. `vdjmatch` instead counts a query's VDJdb neighbours within the ball and compares to the count expected from a matched background control: with $N = |D|$, M the control size and n_C the control neighbour count, $\mathbb{E} = (N/M)\, n_C$ is the expected number of VDJdb neighbours by chance and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D)$ is the enrichment significance — significant only when a clonotype has *more* VDJdb neighbours than the generative background predicts. The Poisson/Chen–Stein theory, the punctured (self-excluding) null and the choice of M are derived in the companion `seqtree` appendix.

First-hit (adaptive) scope.. Rather than fix the ball radius, `vdjmatch` widens the search to each query's *nearest* VDJdb hit — up to 5 total edits with at most 2 insertions and 2 deletions — and

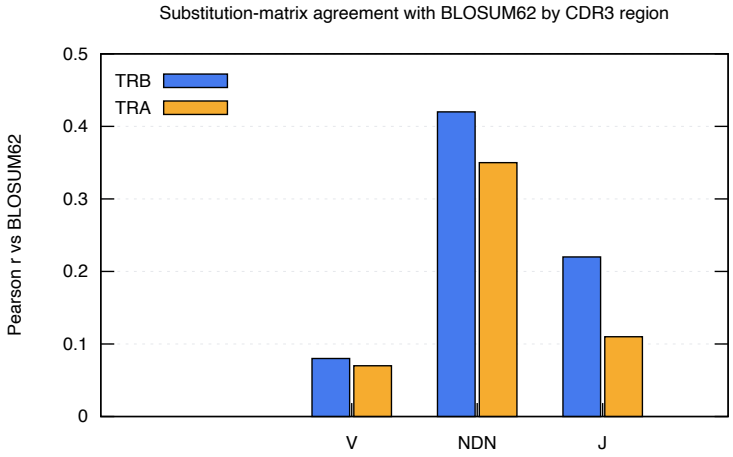


Figure 1: Per-region agreement of the learned CDR3 substitution matrix with BLOSUM62 (Pearson r of off-diagonal scores), human VDJdb. The free-substitution signal lives in the NDN core; the V flank is germline-determined.

evaluates the E-value *at that first-hit radius* r^*. Because the background count $n_C(r)$ grows with r, the calibration adapts per query: a clonotype whose nearest VDJdb neighbour is one substitution away is significant on a single hit (the background ball at $r{=}1$ is essentially empty), whereas one whose nearest hit appears only at $r{=}5$ is not (the background ball is then dense) — so noise is rejected without a hand-tuned scope. A single wide search per side yields the nested counts $n_D(r), n_C(r)$ from the per-hit edit decomposition, and $\mathbb{E} = (N/M)\, n_C(r^*)$, $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\mathbb{E}) \geq n_D(r^*))$ exactly as above (`evalue.first_hit`). The radius may equally be a BLOSUM62+possig *penalty* budget rather than an edit count, defining "nearest" by the position-weighted matrix (§ A.5); we report both. This is the operating point for the cross-method benchmark: on random OLGA repertoires it flags essentially no spurious hits (a generative-background TCR's nearest VDJdb neighbour is far, so $p_{\text{enrich}} \to 1$) while recovering true binders whose nearest labelled neighbour is close.

Paired $\alpha\beta$ E-value.. The null factorization is not an approximation of convenience: α and β chains pair *almost without constraint* in the repertoire at large, as shown by integrating structural and paired-sequencing data [4], so under the background (generative) law the joint ball mass genuinely factorizes, $\pi_0^{\alpha\beta} \approx \pi_0^{\alpha}\pi_0^{\beta}$. Crucially, the *same* analysis finds that *antigen-driven selection distorts* this uniform pairing landscape — i.e. conditioned on an epitope the two chains are *not* independent. This is exactly the structure the joint E-value exploits: the null factorizes (background pairing is unconstrained), so among the N paired VDJdb complexes the expected number of chance joint matches is $\mathbb{E}_{\alpha\beta} = N\,\pi_0^{\alpha}\pi_0^{\beta}$ with p the Poisson tail on the count of complexes matched in *both* chains; an excess over this factorized null is precisely the epitope-conditional dependence (co-selection of α and β) that signals a true specificity. Because the joint null is far smaller than either chain's, true pairs are dramatically more significant: on the paired TCRvdb benchmark, 296/300 pairs match a complex and 295/296 (99.7%) recover the correct epitope, with joint p down to 3.6×10^{-9}. (When the two chains' hits are treated as independent tests, the equivalent statistic is Fisher's combination of the

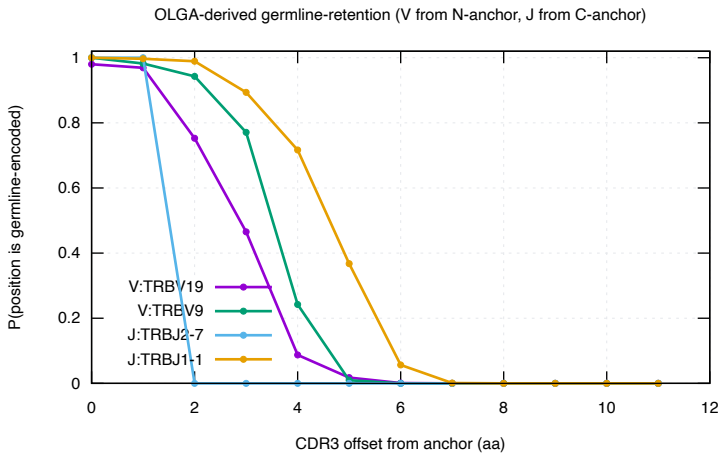


Figure 2: OLGA-derived germline-retention $w(p)$ (via `mirpy`): probability that each CDR3 offset from the V-Cys (or J) anchor is germline-encoded. It falls monotonically into the insert core, defining the region weight $1 - w(p)$ used in scoring.

per-chain enrichments; the appendix `seqtree` co-occupancy term quantifies the residual dependence.)

A.4. Results: does VDJAM (and region weighting) help?

A matrix derived from same-antigen pairs must be tested on epitopes it never saw, or the evaluation is circular. For each held-out epitope E^* we re-derive the NDN matrix from *all other* epitopes, fix the candidate set with a single unit-cost search on E^* (exact self excluded), and re-score each (query, hit) pair four ways — unit, BLOSUM62, NDN-VDJAM, and region-weighted NDN-VDJAM — ranking by score and reporting micro PR-AUC for "hit shares E^*" (Fig. 3, eight largest TRB epitopes). Two honest findings. (i) *Region weighting is a real, consistent, but small improvement over the flat matrix* ($+0.004$ at scope 2, $+0.007$ at scope 3; never worse on any epitope, and the gain grows with scope) — modest because CDR3 variation is *already* NDN-concentrated, so down-weighting the rarely-varying germline flanks moves few hits; it should matter more for cross-V-gene comparisons, where flank (germline-allele) differences are common. (ii) *BLOSUM62 remains the strongest substitution matrix at this narrow-scope retrieval task* (mean PR-AUC 0.32 vs 0.28 region-VDJAM vs 0.29 unit): the data-derived NDN matrix captures general substitution tolerance that BLOSUM already encodes well, and with limited same-antigen pairs it is noisier. At scope 2 the search ball supplies most of the specific-vs-nonspecific discrimination and any matrix only re-ranks within it; the substitution model is therefore a second-order lever here, and the first-order signal is the control-calibrated neighbour count (§ A.3).

A.5. Edit distance, position, and the matrix comparison

Four analyses (`bench/scoring_analysis.py`, `bench/loo_vdjam.py`) settle what scoring actually buys on VDJdb. All figures and numbers below are the *human TRB* example case (the largest, best-

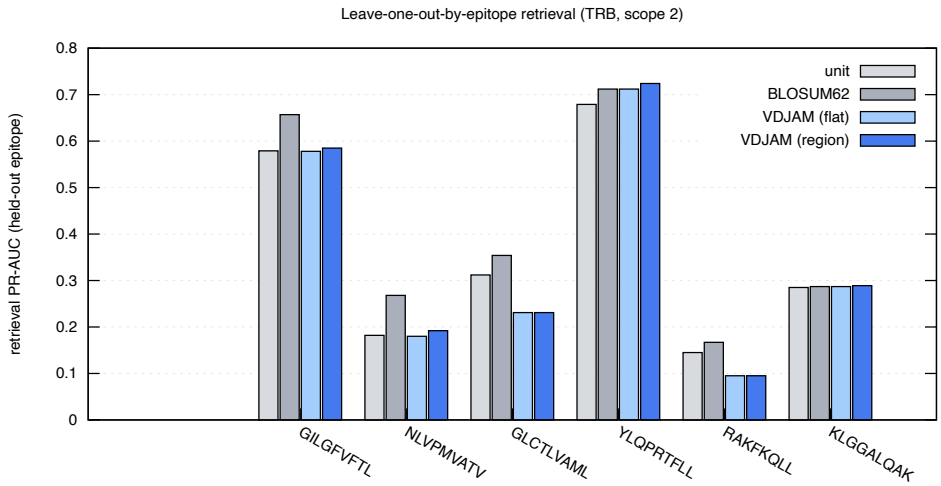


Figure 3: Leave-one-out-by-epitope retrieval PR-AUC (TRB, scope 2): the NDN VDJAM matrix versus BLOSUM62 and unit cost on epitopes excluded from matrix estimation.

populated slice of VDJdb); the same pipeline applies unchanged to human/mouse and to TRA/TRB — point `$VDJDB_SAMPLE`/`$VDJDB_SPECIES` at another export to regenerate.

Hamming distance 1 is the signal:noise optimum.. The fraction of CDR3 neighbours that share an epitope — the precision of a similarity call — falls steeply with edit distance: macro purity (per-epitope mean, robust to the 100× epitope-size range; § below) 0.49, 0.24, 0.15, 0.10, 0.07 at distance $1 \ldots 5$, an enrichment of $56\times \rightarrow 2.5\times$ over chance (Fig. 5). A distance-1 neighbour is twice as likely to be a true co-specific as a distance-2 one and seven times a distance-5 one. This reproduces the original VDJdb observation [6] that single-mismatch neighbours are the sweet spot — and on the dense 2026 release it survives only once a few 10x mega-studies are capped and the random tail is treated as an admixed control (§ A.3); the naive pooled rate reads a near-constant $\sim$0.9 because a random clonotype is probably from a 30k-clonotype epitope. It frames everything below: the search ball, not the substitution matrix, is the dominant lever; the matrix only re-ranks within a ball whose purity is already set by its radius.

Central substitutions carry the specificity signal.. Conditioning on *where* a single substitution falls (Fig. 6), the probability that the two CDR3s still share an epitope dips to ≈ 0.31 in the NDN core but rises to ≈ 0.75 within a few residues of either the V or J anchor: a central mismatch most often *changes* specificity (it is contact-proximal and informative), while a near-anchor mismatch is usually germline noise (the conserved V/J stems). This is the empirical basis for weighting the centre more — the same direction as the germline-retention weight $1 - w(p)$ (§ A.2) — and, encoded directly as a positional weight on BLOSUM62 (the PSSM of § A.5), it is the one change that clearly improves retrieval over flat BLOSUM62 (7/8 epitopes; Table 1). It also matches the contacting-region picture of Shcherbinin et al. [5].

The NDN core is glycine-rich.. Central CDR3 positions reach $\sim$ 31–34% glycine in both chains (Fig. 7) — the N-region/D-gene insertion signature — a strongly non-BLOSUM composition.

Notably TRA is comparably central-G-rich to TRB, so this is an insertion-diversity feature, not a TRB-D-gene-exclusive one; either way the NDN alphabet is distinctive enough that a general matrix is a poor fit there in principle.

No *amino-acid* matrix clearly beats BLOSUM62 — and a genetic-code null ties it.. Re-scoring the held-out-epitope candidates (leave-one-out, § above) with each substitution matrix, scored by balanced PR-AUC on the composition-controlled long list, gives a consistent verdict at every scope (Table 1, Fig. 4): BLOSUM62, PAM250 and the structural matrix are essentially tied at the top, plain Hamming (unit) is close behind, and the data-derived NDN-VDJAM — flat or region-weighted — trails them all. Strikingly, **VDJAMr**, a matrix built from the *genetic code alone* (`loo_vdjam.codon_dissim`: $M(a,b) = \Pr[\text{a single random nucleotide substitution of a random codon of } a \text{ yields } b]$, with no chemistry or selection), *matches* BLOSUM62 ($+0.017$ at distance ≤ 2, $+0.018$ at ≤ 4; ahead in 5–6/8 epitopes) and clearly beats the data-derived VDJAM. That a pure codon-mutational-accessibility null does as well as a chemically tuned matrix says the substitution structure of co-specific CDR3s is largely *generative* (what the recombination/mutation process reaches cheaply), not chemical — which is exactly why hand-tuning the amino-acid alphabet does not help. The substitution matrix is a second-order lever; the first-order statistic remains the control-calibrated neighbour count at distance 1 (§ A.3).

But position-weighting BLOSUM62 does (the significance PSSM).. Experiment 2 says *where* a substitution falls matters more than *which* substitution it is. We encode that as a positional factor on top of BLOSUM62. Crucially the profile is **anchored from both ends**, not by relative position: the germline V/J stems sit at fixed *absolute* offsets from the CDR3 termini regardless of length, so we learn two end-anchored profiles Pr(same | sub at offset d from the V anchor) and the J-anchor counterpart (Fig. 6), Beta-Binomial-smoothed so sparse deep-core offsets shrink toward the global rate (`regions.load_significance`; estimated by `bench/scoring_analysis.py`). The per-position weight is $\omega(p) = 1 - \max(\Pr_V(d_V), \Pr_J(d_J))$ with $d_V = p,\ d_J = L-1-p$, normalised to mean 1: a position is down-weighted if it lies near *either* anchor, up-weighted in the NDN core. This is the data-learned twin of the germline-retention weight (both $1 - \max(\text{V-side}, \text{J-side})$). `seqtree` consumes it natively — `regions.significance_pssm()` builds `PositionalMatrix.from_weights(BLOSUM62, [100 ω(p)])`, so $\text{pen}(p,a,b) = \omega(p) \cdot \text{BLOSUM62}(a,b)$. The multiplicative form is justified by the data: stratifying single substitutions by anchor offset $\times$ BLOSUM severity, substitution severity matters *only in the core* (Pr(same) 0.50 conservative vs 0.38 radical) and is flat near the anchors — BLOSUM discriminates exactly where ω is large. This is the only scoring change that clearly beats flat BLOSUM62, winning 7/8 held-out epitopes at distance ≤ 2 and 6/8 at ≤ 4 (Table 1); the gain is uniformly signed because the matrix re-ranks within a ball whose purity is already set by the radius. The remaining lever, the *V-gene* dimension, is taken up in § A.6.

A purpose-built TCR matrix (tcrBLOSUM) does not transfer.. Postovskaya et al. [3] report that tcrBLOSUM — a BLOSUM-style matrix counted from same-epitope, equal-length CDR3 pairs — improves epitope-specific TCR clustering over BLOSUM62. Two features make that gain suspect: the counts apply *no* redundancy clustering (so convergent public clonotypes dominate the statistics), and the matrix is validated on the same same-epitope relation it is fit to. Run through our leave-one-out-by-epitope retrieval (their published tcrBLOSUM_b, `bench/tcrblosum_refute.py`), it does *not*

beat BLOSUM62: 0.555 vs 0.567 at distance ≤ 2 (ahead in 2/8 epitopes) and essentially tied at ≤ 4 (0.589 vs 0.591), while their bundled BLOSUM62 reproduces ours exactly (a loader sanity check). Reweighting *either* matrix by the central-significance PSSM helps, and it helps BLOSUM62 the most (BLOSUM+possig 0.603 vs tcrBLOSUM+possig 0.598). The lesson is consistent: a re-counted amino-acid alphabet does not transfer across epitopes; position does.

balanced PR-AUC	unit	BLOSUM62	PAM250	structural	VDJAM	VDJAMr	VDJAM-region	BLOSUM+possig
edit distance ≤ 2	0.543	0.564	0.572	0.564	0.524	0.581	0.529	**0.598**
edit distance ≤ 4	0.573	0.585	0.590	0.585	0.535	0.603	0.543	**0.611**

Table 1: Substitution matrices by leave-one-out-by-epitope VDJdb-vs-VDJdb retrieval (human TRB, 8 largest epitopes of the 2026-06-11-ZENODO composition-controlled long list; balanced PR-AUC). The chemical matrices (BLOSUM62/PAM250/structural) and the genetic-code null VDJAMr are all ≈tied; the data-derived VDJAM trails; only BLOSUM62 reweighted by the positional-significance PSSM clearly leads (7/8 and 6/8 epitopes).

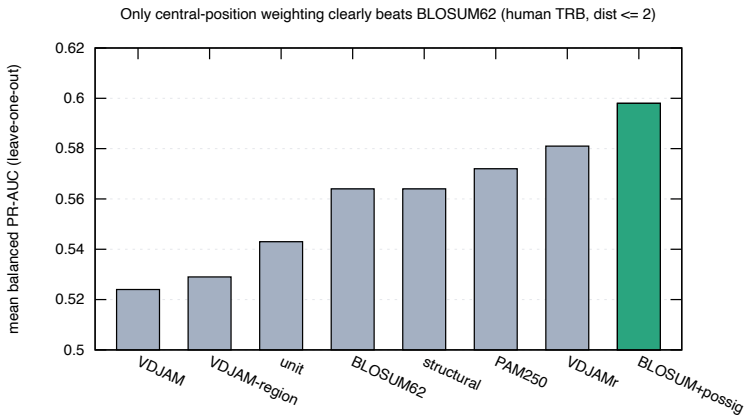


Figure 4: Mean leave-one-out retrieval PR-AUC by scoring (TRB, distance ≤ 2). No standard amino-acid matrix beats BLOSUM62; reweighting BLOSUM62 by the central-position significance factor (green) does.

A.6. The V-gene dimension

The remaining lever is the V gene. Three questions on the composition-controlled long list (human TRB unless noted): is a V match itself a specificity signal, does it soften to germline-similar V genes, and — if so — does it localise to a few contacting-loop positions (a V "pseudosequence")?

A V match is a strong, near-binary specificity prior.. Splitting every scored neighbour pair by whether query and hit use the same V family, same-V neighbours share the held-out epitope 44% of the time versus 6% for cross-V (edit distance ≤ 4; Table 2) — a $\sim 7\times$ prior, far larger than any substitution-matrix effect. This is why the legacy tool used V as a hard filter, and it argues for carrying V as a strong prior (a soft $S(V)$ term or a per-V null) in a joint V+CDR3 score. Position

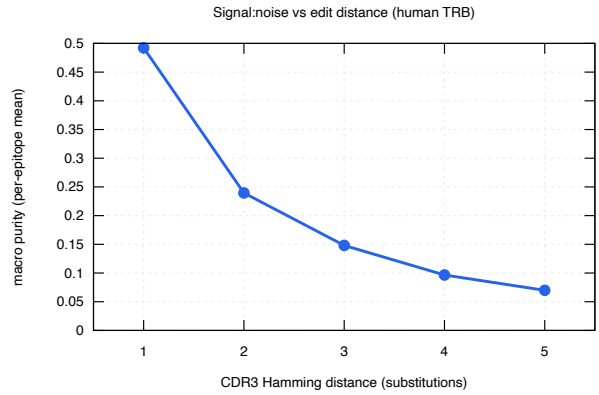


Figure 5: Macro purity (per-epitope mean fraction of CDR3 neighbours sharing an epitope) vs Hamming distance (human TRB, composition-controlled long list): distance 1 is the signal:noise optimum [6].

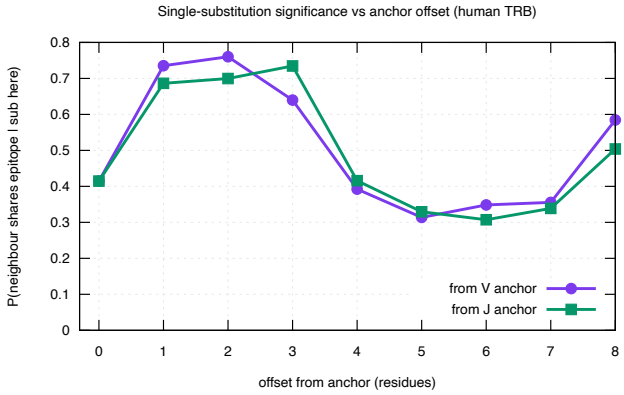


Figure 6: Pr(neighbour shares epitope | single substitution at relative position), human TRB. The NDN core is where a mismatch most changes specificity; the V/J borders are germline noise.

weighting sharpens the *same*-V stratum most (balanced PR 0.632 → 0.713 with the significance PSSM, +0.081) — where the germline frame is shared and only the NDN core varies — and little in the noise-dominated cross-V stratum (+0.012).

Coarse germline-loop similarity barely predicts it.. If the same-V advantage were simply the chemistry of the V-encoded contacts, cross-V co-specificity should rise smoothly with germline CDR1+CDR2 similarity. As a *continuous* quantity it barely does: across every {human,mouse}×{TRA,TRB}×{MHC-I,MHC-II} cell (`bench/vgene_scan.py`, Fig. 8, Table 4) the V prior is universal (same/cross ratio 1.0–8×) but the point-biserial correlation between loop similarity and co-specificity is ≈ 0 ($r \in [-0.06, +0.06]$ in the well-powered cells). Binning cross-V pairs by similarity shows only a weak rise — 6% at low similarity to 12% at the top bin, still far below the same-V 44% (Table 3). So a loose, continuous "fuzzy-V" similarity does not recover the prior.

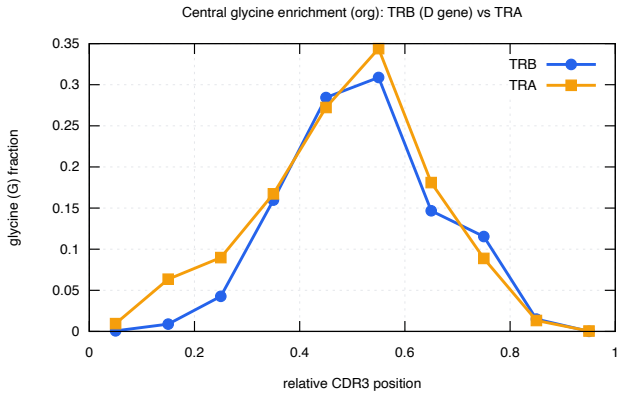


Figure 7: Glycine fraction by relative CDR3 position (human): the NDN core is ∼ 30–36% glycine in both TRA and TRB (insertion/D-gene signature).

Near-*exact* germline identity partly recovers it — at CDR3-like tolerance.. Resolving the similarity to exact germline-region matches changes the picture (`bench/vregion_decompose.py`, human TRB MHC-I; Table 5). Cross-V co-specificity rises *monotonically* as the germline CDR1+CDR2 edit distance shrinks — 10.5% at ≥ 6 mismatches, 12.6% at 3–5, 17.9% at 1–2, and 32% at 0 (identical loops): a real ∼3× lift over the cross-V floor. FR1 identity behaves likewise (27.6%, 2.6×). So the germline contacting loops *do* carry part of the V signal, at a tight CDR3-like tolerance (near-exact identity; the continuous metric above, dominated by 1–2-mismatch pairs, misses it) — but even identical loops recover only about *half* the same-V advantage (32% vs 64%); the rest is gene-identity-specific (the exact CDR3-germline anchor and integrated docking geometry). Caveat: the identical-loop bin is small ($n = 25$ at distance ≤ 2, $n = 5$ at ≤ 1, and bounced 32–60% across held-out sets) — the robust claim is the monotone gradient over the well-powered bins ($n = 507$, 3735, 36853), not the exact endpoint.

The recovery is whole-loop, not a sparse pseudosequence.. Do a few decisive positions carry it (the MHC-pseudosequence analogue)? Per-position residue-match lift over the cross-V baseline (`bench/vpseudo.py`, length-constant CDR1/CDR2) is mostly flat (lift 1.0–1.2 at the 10.5% baseline), with a single exception — CDR2 position 5 (1.56×, $n = 4833$). So the partial recovery above is *whole-loop* near-identity, not the matching of a handful of pseudosequence coordinates: a different V gene must reproduce essentially the entire CDR1+CDR2 to behave like the same V. A useful soft-V match must therefore demand near-exact loop identity, not loose similarity. The next step is the full IMGT-numbered treatment (FR positions and the FR3 DE-loop/"CDR2.5", via the `arda` IMGT/V-QUEST milestone) to test whether specific positions sharpen this.

A.7. Benchmark: the reference-replicated shortlist

The latest VDJdb release is our standing benchmark, but its labels vary in confidence. The cleanest gold standard is a clonotype–epitope association reported independently in ≥ 2 references (`db.replicated`, null references excluded; ∼3000 TRB and ∼2000 TRA such pairs in the current human release) —

V stratum	pairs	co-specific	flat	+possig	+region
same V	120 738	45.2%	0.626	0.682	0.678
cross V	979 199	6.2%	0.558	0.569	0.538

Table 2: V-gene-stratified leave-one-out retrieval (human TRB, edit distance ≤ 4; co-specific = positive rate, the rest balanced PR-AUC). A V match is a $\sim 7\times$ co-specificity prior; the significance PSSM helps the same-V stratum most (+0.056).

germline CDR1+CDR2 similarity	cross-V pairs	co-specific
[0.0, 0.3)	479 181	5.8%
[0.3, 0.5)	387 839	6.2%
[0.5, 0.7)	89 533	7.4%
[0.7, 0.9)	18 320	10.2%
[0.9, 1.0)	4 326	11.6%

Table 3: Continuous germline CDR1+CDR2 similarity gives only a weak rise in cross-V co-specificity (6% → 12%), far below the same-V 45% — which is why $r \approx 0$. The stronger signal lives at near-*exact* identity (Table 5), not along the continuous similarity axis.

the same "shortlist" idea used for `mhcmatch`. We measure annotation accuracy on it by leave-one-out nearest-neighbour voting against the rest of VDJdb at the signal:noise-optimal scope (subs 1), excluding the clonotype's own exact CDR3 so the label must come from *other* TCRs (`bench/shortlist_accuracy.py`). Replicated TCRs are annotated at $\sim$44–47% top-1, modestly above size-matched single-reference controls of the same epitopes (Table 6). The dramatic $3\times$ gap we saw on a sparse older export was an artefact: on the dense 2026 release even single-reference clonotypes have findable neighbours (control recall $\sim$48–72%), so the gold set is cleaner but no longer dramatically *easier*. Restricting neighbours to the same V family helps at the wider subs 2 scope (TRB 53.1 → 56.8%) — exactly where cross-V false neighbours appear (§ A.6), confirming the V prior is worth its place.

A.8. Worked examples

Pairwise (different lengths).. `seqtree` aligns CDR3s of differing length, placing indels where the loop genuinely lengthens. The edit-distance alignments below (influenza GIL-specific TRB CDR3s) show the conserved `CASS` V-anchor and `DTQYF` J-motif with substitutions and indels confined to the central NDN core:

```
query (ref): CASSPRSGETQYF   epitope GILGFVFTL   (unit / edit-distance alignment)

vs CASSNRSGETQYF  (len 13; subs=1 ins=0 dels=0; CIGAR 4=1X8=)
  CASSPRSGETQYF
  |||| ||||||||
  CASSNRSGETQYF

vs CASSIRSGESTQYF  (len 14; subs=1 ins=1 dels=0; CIGAR 4=1X4=1I4=)
  CASSPRSGE-TQYF
  |||| |||| ||||
```

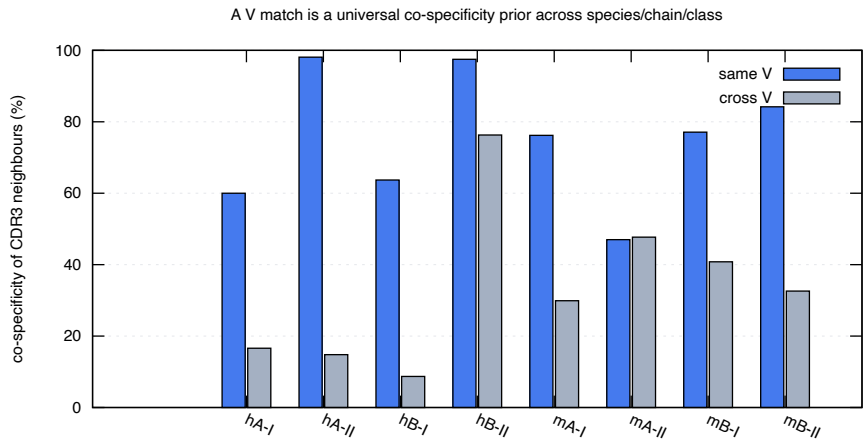


Figure 8: Same-V vs cross-V CDR3-neighbour co-specificity across species/chain/MHC-class cells of VDJdb (h/m = human/mouse; A/B = TRA/TRB; I/II = MHC class). A V match is a universal co-specificity prior; continuous CDR1/CDR2 similarity does not bridge the gap (Table 4).

cell (species / chain / class)	same-V	cross-V	ratio	r(vsim, co-spec)
human TRA MHC-I	60.0%	16.6%	3.6	0.050
human TRA MHC-II	98.1%	14.8%	6.6	0.269
human TRB MHC-I	63.7%	8.7%	7.3	0.070
human TRB MHC-II	97.5%	76.3%	1.3	0.054
mouse TRA MHC-I	75.1%	29.3%	2.6	0.021
mouse TRA MHC-II	47.0%	47.7%	1.0	−0.009
mouse TRB MHC-I	77.1%	40.8%	1.9	−0.080
mouse TRB MHC-II	84.2%	32.6%	2.6	−0.017

Table 4: Per-cell same-V vs cross-V co-specificity and the point-biserial correlation between continuous germline CDR1+CDR2 similarity and co-specificity among cross-V pairs (top 6 epitopes/cell, ≥ 30 CDR3/epitope). The V prior holds everywhere (ratio 1–8×); continuous loop similarity barely predicts ($r \approx 0$; the 0.24 is human TRA MHC-II, whose high-similarity bin has $n = 4$).

```
  CASSIRSGESTQYF

vs CASSLFSETQYF  (len 12; subs=2 ins=0 dels=1; CIGAR 4=2X1=1D5=)
  CASSPRSGETQYF
  ||||  | |||||
  CASSLFS-ETQYF

vs CASSLGETQYF  (len 11; subs=1 ins=0 dels=2; CIGAR 4=2D1X6=)
  CASSPRSGETQYF
  ||||   ||||||
  CASS--LGETQYF

vs CASSPWSGSQETQYF  (len 15; subs=1 ins=2 dels=0; CIGAR 5=1X2=2I5=)
```

cross-V stratum (human TRB MHC-I)	co-specific	n
cross-V baseline	10.8%	41 132
CDR1+CDR2 edit distance ≥ 6	10.5%	36 853
edit 3–5	12.6%	3 735
edit 1–2	17.9%	507
edit 0 (identical)	**32.0%**	25
FR1 identical	27.6%	257
same-V reference	63.8%	14 883

Table 5: Cross-V co-specificity rises monotonically as germline CDR1+CDR2 approach identity; at exact identity it reaches ∼32% — a real ∼3× lift over the cross-V floor, but only about *half* the same-V level (64%). Germline-loop identity carries part of the V prior at a tight, near-exact tolerance; the rest is gene-identity-specific. (edit 0/CDR1&CDR2 bin small, $n = 25$.)

chain	set	n	top-1 (CDR3)	top-1 (V-restr.)	recall
TRA	shortlist (≥ 2 ref)	1993	43.9%	43.8%	62.3%
TRA	control (1 ref)	1993	33.0%	29.6%	47.5%
TRB	shortlist (≥ 2 ref)	2962	47.3%	47.1%	54.2%
TRB	control (1 ref)	2962	46.7%	43.7%	50.2%

Table 6: Leave-one-out epitope-annotation accuracy on the reference-replicated shortlist vs a size-matched single-reference control (human, subs 1, exact self excluded). Top-1 = nearest-neighbour-vote epitope correct; recall = true epitope reachable within scope. On the dense 2026 release the shortlist is modestly above control, not the 3× of the older sparse export.

```
CASSPRSG--ETQYF
|||||  ||  |||||
CASSPWSGSQETQYF
```

Multiple (same length).. Aligning same-length same-epitope CDR3s (no indels needed) makes the conservation profile explicit (Figs. 9–10): columns are shaded by conservation, dark at the germline-determined CASS prefix and J motif, light through the variable NDN core. This is the structural counterpart of the finding of Shcherbinin et al. [5] that specificity-preserving CDR3 substitutions occur predominantly outside the peptide/MHC-contacting positions while preserving the loop backbone — i.e. the tolerated variation we score in the NDN core is exactly the variation that does not break recognition.

A.9. Limitations and roadmap

The substitution *alphabet* is a second-order lever: the data-derived NDN-VDJAM trails BLOSUM62 at every scope, and a chemistry-free genetic-code null (VDJAMr) does just as well as BLOSUM62 — the co-specific-CDR3 substitution structure is largely generative, not chemical (§ A.5). What moves the needle is *position*: reweighting BLOSUM62 by the central-significance PSSM clearly beats flat BLOSUM62 (7/8 epitopes), confirming that for CDR3 *where* a mismatch falls matters more

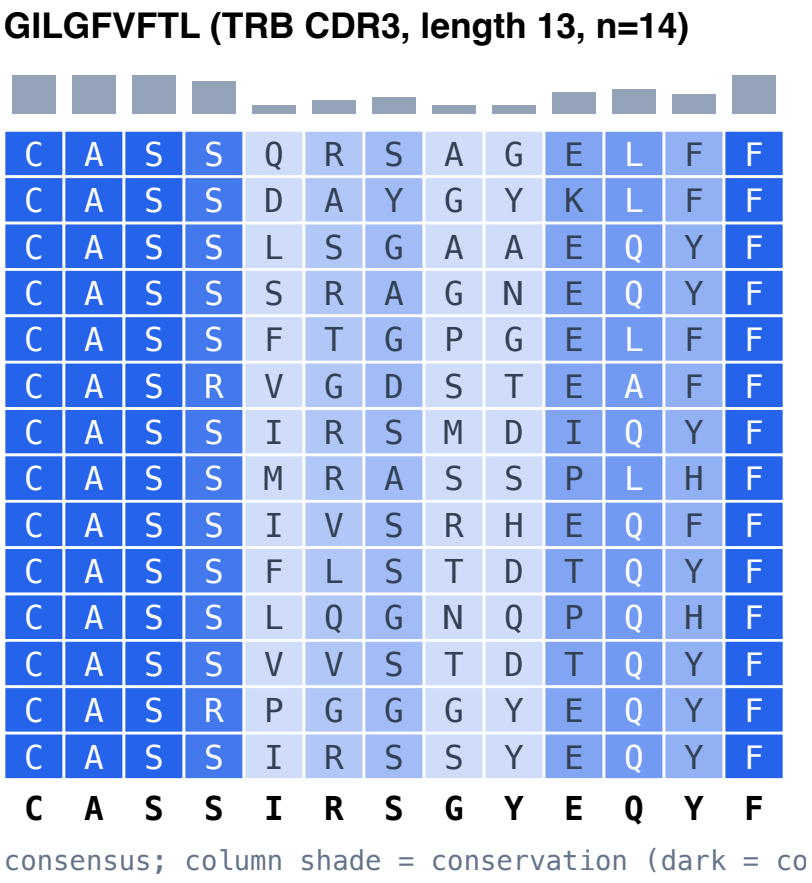


Figure 9: GIL (influenza M1, A*02) TRB CDR3 block; conserved V/J flanks, variable NDN core.

than *which* residue it is. The *V-gene* dimension (§A.6) is a strong near-binary prior (∼7×); it is *not* captured by loose germline-loop similarity, but cross-V co-specificity does rise at *near-exact* CDR1+CDR2 identity — recovering about *half* the same-V advantage (∼32% vs 64%) at a tight, CDR3-like tolerance, whole-loop rather than via a sparse pseudosequence (exact-match bin small, $n \approx 25$). The open work is to model V as a strong prior in a joint V+CDR3 E-value, to demand near-exact (not loose) loop identity for any soft-V match, and to extend the position analysis to full IMGT numbering (FR positions and the FR3 DE-loop, via the `arda` IMGT/V-QUEST milestone); a per-chain treatment (TRA vs TRB) is also pending. Separately, the control's V–J usage must match the query repertoire's or the null is mis-specified — whether to renormalize V–J usage (at the risk of erasing genuine V-gene bias) is an open question to settle on held-out epitopes.

Support, not the score, decides the linked-pair metric.. The distance-local rank statistic of the companion appendix (`score-derivation`) averages a normalised reciprocal background rank over all N references of a target, imputing references outside the computed radius at their null-conditional mean. That imputation is unbiased but uninformative, and when *no* reference lies inside the radius the statistic reduces to a function of the control count alone, carrying no information about the target. The practical consequence is a clean monotone dependence on target support: measured over 620

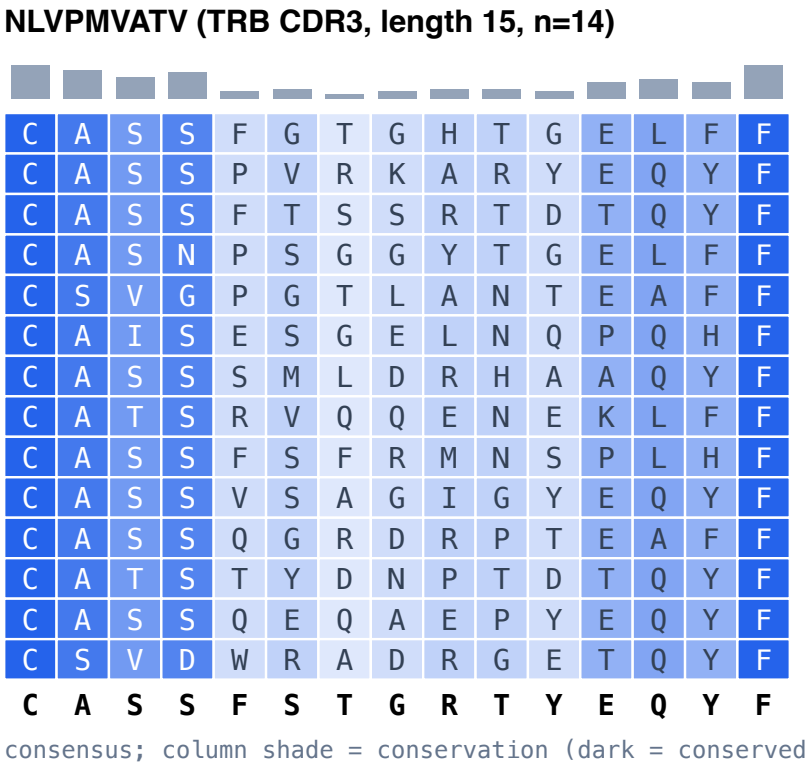


Figure 10: NLV (CMV pp65, A*02) TRB CDR3 block; same conservation pattern.

benchmark tasks, the mean average-precision difference against the original scorer is —0.064 for targets with 30–100 positives, —0.024 at 100–300, +0.007 at 300–1000 and +0.010 above 1000, and the same monotone pattern holds for *every* distance-geometry arm we tested, including the unmodified native TCRdist3 comparator. Restricting each task to the queries that do have multi-reference support removes about half the low-support deficit, reverses the sign above 300 positives, and removes the entire linked-pair deficit (—0.141 → +0.001 over 86 paired tasks). The effect is largest for linked pairs because the paired route applies the same numeric radius to the *sum* of the two chain distances — a roughly twofold tighter per-chain constraint — so the median fraction of queries with a supported linked reference is 0.17, against 0.82 for single chains. On well-populated targets, when a linked reference pair *is* inside the radius, both the original scorer and the rank statistic reach average precision 0.998–1.000. The open work is therefore to raise paired coverage and to give the zero-support case an explicit "no multi-reference evidence" disposition rather than a number, not to tune the score; and to separate the linked-pair radius from the single-chain radius, which are different quantities that currently share a numeral. The disposition change alone, leaving the statistic untouched, raises mean average precision from 0.818 to 0.846 over all 99 linked-pair tasks (69 wins, 23 losses, 7 ties); recomputing the statistic over supported references only instead *loses* ground, because the censored far-reference mass is genuinely informative once most references are far. Support is itself a sharp indicator here — on well-populated targets a linked reference pair lies within the current radius for 0.59 of positives against 0.06 of negatives — which is why ordering

by support recovers so much for pairs and almost nothing for single chains, where support is near-universal. That specificity comes from requiring a *common* reference pair: asking only that each chain independently have some reference neighbour covers 0.97 of positives and 0.95 of negatives, a purity ratio of 1.0 against 11.1, so the tight summed constraint is the source of paired discrimination rather than an obstacle to it. The remaining lever is then routing rather than scoring. Most linked-pair queries — 78% overall — have no common reference pair but do have per-chain neighbours; ordering those by a scale-free fusion of both chains' single-chain evidence, while leaving the single-chain route untouched, raises pooled mean average precision over 625 benchmark tasks from 0.821 to 0.829 and the median from 0.841 to 0.848, with linked-pair tasks rising from 0.818 to 0.868. Reformulating the rank statistic itself, by contrast, does not move the metric at all.

A further direction is to predict, *a priori*, which epitopes are intrinsically easy versus hard to annotate — the wide spread of per-epitope PR-AUC in our experiments (e.g. the convergent, "featured" influenza GIL vs the diffuse, "featureless" CMV NLV) is not noise but biology. A pMHC with prominent TCR-facing features selects a focused, motif-rich repertoire that clusters cleanly, whereas a featureless one is recognised by structurally diverse TCRs in varied docking modes and yields no tight motif [8, 9, 7, 2]; clustering-based specificity inference succeeds or fails accordingly [1]. A model that scores epitope "annotability" from repertoire convergence (and, where available, pMHC structural features) would let us report calibrated confidence per epitope and triage NLV-like hard cases — a planned addition to the benchmark. These extensions, and the full comparison against existing tools, are ongoing work.

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